Weaker congruences and even Pell indices

Smaller explicit equations for Con(ZFC)

A research package for Vladimir Reshetnikov's ProveIt project

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Abstract

We give two concrete improvements to the explicit Diophantine consistency equations in a pinned ProveIt snapshot. First, we weaken a divisibility carrier, use the actual coefficient size to resolve its fifth-power congruence, and combine two congruences in one homogeneous intersective polynomial. The resulting equations have twelve unrestricted integer unknowns and exact degrees $768, 784, 800, 816, 1040$, with respectively $161, 154, 147, 144, 140$ binary arithmetic operations. The prior comparable points had degrees 1200 and 1456 with 145 and 141 operations. Second, in the direct C222 construction, we combine signs for repeated bounded intervals and replace power requests $m = n^k$ by $m^2 = n^{2k}$. Even Pell indices eliminate twelve index-sign predicates. The complete result is a quartic in 566 unrestricted integers, with 6431 monomials and coefficient height 2445790974704, compared with 604 variables and 9676 monomials at the same height in the pinned parent. Every equation, constant payload, projection certificate and verification program is supplied. The new reductions have ordinary unbounded proofs; their consistency interpretation composes with the explicitly cited existing source/calculus calibration. No new Lean or Rocq end-to-end proof, unconditional consistency assertion, global optimum, or literature-wide priority claim is made.

1 What is constructed, and what is improved

For each final polynomial P in this package, the reductions below and the cited source calibration give

$$\mathsf{ZFC} \vdash (\neg\exists \mathbf{x} \in \mathbb{Z}^n \; P(\mathbf{x}) = 0) \iff \mathrm{Con}(\mathsf{ZFC}). \tag{1}$$

Here consistency is the usual arithmetic assertion that there is no finite syntactic proof of a contradiction in the fixed calculus. We prove equivalence of *existence* of polynomial zeros to finite refutability. We do not prove that the displayed polynomials have no zeros.

The comparison is to ProveIt commit `84e16e43092614bae262ca59651cb9d77e31a255`, including its active research and incoming reports, rather than only the older collection README [1, 2]. "Simplest" is not a total order: variable count, degree, operation count, coefficient size and proof complexity can conflict. The following are actual, fully materialized points.

Construction	Integer variables	Degree	Operations	Status
Prior scalar carrier, rational	12	1200	145	inherited
Prior scalar carrier, integral	12	1456	141	inherited
Weak carrier, integral	12	1040	140	new
Weak carrier, rational	12	816	144	new
Centered weak carrier	12	800	147	new
Mixed-modulus carrier, integral	12	784	154	new
Mixed-modulus carrier, rational	12	768	161	new

Operations count every live binary addition, subtraction and multiplication, including squares and scalar products. Named polynomial values add no unknowns. All six large coefficient values are unchanged in these comparisons. The 140/1040 and 144/816 points improve both degree and operations relative to the corresponding old points. The lower-degree points make explicit cost/degree tradeoffs.

Direct quartic	Variables	Quadratic rows	Monomials	Height
Pinned parent	604	372	9676	2445790974704
Seven sign combinations	589	371	8157	2445790974704
Even-index compiler	566	372	6431	2445790974704

The last reduction removes 38 variables and 3245 monomials from the matched direct parent, approximately 6.29% and 33.54%, respectively. The direct branch uses C222, not K215 or Jones's universal polynomial. The twelve-variable branch retains the repository's existing native source.

Ten-integer high-degree constructions already exist in the same repository; Sun's general ten-integer theorem is also a relevant benchmark [8]. The present twelve-variable equations do not improve the least known variable count. Their concrete gain is the degree/cost frontier within a completely instantiated family.

2 The complete twelve-variable equation

The only quantified unknowns are

$$(j, C, z, h, r, v, w, T, c, n, t, s) \in \mathbb{Z}^{12}.$$

All other symbols below are polynomial abbreviations. The six fixed constants are $\beta, K_0, M_C, M_F, A_c, B_c$ and are specified in Section 3. Define the source front by

$$J = 6j + 1, \qquad q = \beta J + 1, \qquad W = 12w + 4, \tag{2}$$

$$X = qW, \qquad Y = qv, \qquad a = (X + 1)Y, \tag{3}$$

$$H = 4a + 3, \qquad \Delta = a^2 + H = (a + 2)^2 - 1, \tag{4}$$

$$F = (K_0 + W)C - z(q - 1),$$

$$R = (q(q - F) - C)(q^2 - 1) + (M_C + qM_F)J, \tag{5}$$

$$k = R + 1 + hXY, \qquad \eta = c - Yk, \qquad \zeta = k - \eta. \tag{6}$$

The strict source guards are encoded through

$$G = (A_cF)(B_cC) - (q - A_cF - B_cC)^2, \tag{7}$$

$$E = \eta\zeta, \tag{8}$$

$$B_0 = GE - (t^2 - G - E + 1)^2, \tag{9}$$

$$D_g = (3B_0 - 4)s^2 + 1. \tag{10}$$

The Pell registers are

$$I = src^2, \qquad D_f = 1 + \Delta I^2, \qquad Q = \Delta^2 I^2, \tag{11}$$

$$\delta = Q - 1, \qquad V = D_f(c - R + 4IT) - c, \qquad C_y = QV^2 - 1, \tag{12}$$

$$\ell = XY^2, \qquad D_\tau = 1 + \ell(\ell + 1)k^2. \tag{13}$$

Finally define the new carrier:

$$K_* = X(2X - 3c) - 2, \qquad M = 2q^2, \qquad D = MH, \tag{14}$$

$$C_* = MK_* + H(v - q^2), \qquad z_* = DT + C_*, \tag{15}$$

$$A_* = (z_*^2 + 3D^2)(z_*^3 - 19D^3), \qquad S = srM^5H. \tag{16}$$

Definition 2.1 (Four-root norm)**.** Let

$$\mathcal{N}_4(x;a,b,e,d)=\prod_{\varepsilon\in\{\pm1\}^4}(x-\varepsilon_1\sqrt{a}-\varepsilon_2\sqrt{b}-\varepsilon_3\sqrt{e}-\varepsilon_4\sqrt{d}). \tag{17}$$

This is an ordinary integer polynomial in x,a,b,e,d: changing the sign of any formal square root permutes the factors. Its degree in x is 16, and its degree in each radicand is 8.

Our main equation is

$$\boxed{\begin{aligned}P_{768}=\delta^8\mathcal{N}_4(S(n+V)-A_*;\\ 9S^2D_f,\ 81S^2D_\tau,\ 729S^2D_g,\ S^2C_y/\delta)=0.\end{aligned}} \tag{18}$$

The division in this display is only a compact presentation. Expanding the polynomial norm and multiplying by δ^8 clears every denominator. The supplied 161-operation DAG contains no division, radical, exponent with a variable exponent, or existential subroutine. Appendix A gives a division-free evaluator.

The companion equation is

$$\boxed{\begin{aligned}P_{784}=\mathcal{N}_4(S(n+\delta V)-A_*;\\ 9S^2D_f,\ 81S^2D_\tau,\ 729S^2D_g,\ S^2\delta C_y)=0.\end{aligned}} \tag{19}$$

It has 154 operations and degree 784. Both domains are unrestricted integers. The file names `rational.json` and `integral.json` refer to their auxiliary presentations, not to the domains of the solutions.

3 Fixed constants and the inherited logical endpoint

The six constants are the complete source34 tuple authenticated in the pinned scalar-carrier packet [3]. Three have short closed forms:

$$\beta=2^{1064921}-1,\qquad A_c=2^{1113},\qquad B_c=2^{666104}.$$

The other three are supplied as complete, minimally encoded little-endian nonnegative integers in the compressed payloads. Their bit lengths are

Constant	Bit length	Role
β	1064921	source base minus one
K_0	1065043	fixed source stencil
M_C	1064921	source mask
M_F	1064922	already shifted source mask
A_c	1114	determinant scale
B_c	666105	determinant scale

The authoritative binding file gives their byte counts and SHA-256 digests; the verifier decompresses and authenticates every byte. These are fixed numerals, not extra variables or consistency-dependent choices. No port from a different, smaller coefficient tuple is substituted.

The inherited endpoint is the signed native/Pell source theorem for this exact tuple. Its completed ordinary argument identifies source witnesses with finite refutations, through the pinned native compiler and controller calibration. The local proofs retained here are `PRIME_TWELVE.md`, `INTERPOLATION12.md`, `signed-quadratic-proof.md`, `RESIDUE_AND_DETERMINANT.md`, and the scalar-parent proof [3, 4, 5]. The source theorem is used as an explicit earlier theorem, just as a construction might use a published Diophantine representation theorem. The current work does not replace its ordinary proof by a new proof-assistant derivation.

The new contribution must therefore be read precisely: the equations are fully instantiated, and the new unbounded reductions are proved below; the logical interpretation composes those reductions with the cited existing calibration. Finite circuit tests alone do not certify that calibration and do not establish (1) on their own.

4 Why a weaker carrier is sufficient

The old scalar carrier was designed to force $H \mid K$ directly. But its later use is to show that a certain explicit integer B vanishes by combining divisibility with a size estimate. That last step admits a weaker hypothesis.

Lemma 4.1 (Power congruence with a height bound)**.** *For nonzero integers H and positive integers d, if $H \mid B^d$ and $|B|^d < |H|$, then $B = 0$.*

Proof. The integer B^d/H has absolute value less than one, so is zero. □

The lemma is elementary. Its useful role here is to change what the upstream carrier must certify, while retaining the final exact conclusion. This is a reduction in the equation, not an approximate replacement of a logical condition.

Put

$$\pi(u) = (u^2+3)(u^3-19), \qquad \Phi_D(z) = D^5\pi(z/D) = z^5 + 3D^2z^3 - 19D^3z^2 - 57D^5. \tag{20}$$

The retained intersectivity proof establishes a root of π modulo every positive integer, including powers of $2, 3, 19$. The polynomial has no rational root: $u^2 + 3 > 0$ over the reals and 19 is not a rational cube.

Lemma 4.2 (Two congruences in one carrier)**.** *Let M, H be nonzero coprime integers, $D = MH$, and*

$$C_* = MK_* + HL, \qquad z_* = DT + C_*.$$

If $M^5H \mid \Phi_D(z_*)$, then $M \mid L$ and $H \mid K_*^5$.

Proof. For $p^e \| M$, suppose $b = v_p(z_*) < e$. In (20), the leading term has valuation $5b$. All other terms have valuations at least $2e + 3b$, $3e + 2b$, and $5e$, strictly larger. Thus the sum has valuation $5b < 5e$, contradicting $M^5 \mid \Phi_D(z_*)$. Hence $M \mid z_*$ and, by coprimality, $M \mid L$. Reducing the carrier modulo H gives $H \mid (MK_*)^5$, and coprimality cancels M. □

The difference in strength is intentional: the M part is recovered exactly, while only a fifth-power congruence is required at H. This allows the inexpensive source scale $Y = qv$ in (4); its original cubic restriction is recovered from $M \mid v - q^2$.

5 Soundness, with all exceptional branches in order

5.1 Uniform arithmetic facts

The actual β is odd and is 1 modulo 3. Thus q is even, $q \equiv 2 \pmod 3$, and $q \neq 0$. Also $W \equiv 4 \pmod{12}$, so

$$|X| \geq 4(\beta - 1) > 2^{36}, \qquad X \equiv -1 \pmod 3. \tag{21}$$

If $v = 0$, then $a = 0, H = \Delta = 3$. Otherwise $|a| = |qv(X + 1)| > 1000$. Therefore $\Delta \geq 3$, H is nonzero and odd, and Q is either zero or at least 9. In particular δ never vanishes. Since a is divisible by q,

$$H \equiv 3 \pmod q, \qquad \gcd(H, M) = \gcd(H, 2q^2) = 1.$$

Modulo 3 one has

$$a = H = \Delta = 0, \quad D_f = D_\tau = 1, \quad Q = 0, \quad \delta = C_y = -1. \tag{22}$$

For the first Pell radicand, $\ell = XY^2$ is 0 or -1 modulo 3. Uniformly $D_f, D_\tau \geq 1$, since $\ell(\ell + 1) \geq 0$ for every integer ℓ. No sign of Y or ℓ has been assumed.

5.2 Nonzero scale, square extraction and the strict guard

If $S = 0$, then $sr = 0$ and hence $I = 0$. Every scaled radicand vanishes; both equations reduce to A_*^{16}. But

$$A_* = D^5\pi(T + C_*/D) \neq 0$$

because $D \neq 0$ and the argument is rational. Thus every zero has $s \neq 0$ and $r \neq 0$.

Divide a vanishing conjugate factor by S. For P_{768} the four root terms are

$$\sqrt{C_y/\delta}, \quad 3\sqrt{D_f}, \quad 9\sqrt{D_\tau}, \quad 27\sqrt{D_g}.$$

For P_{784} the first term is $\sqrt{\delta C_y}$. Their squared 3-adic orders are $0, 2, 4, 6 + v_3(D_g)$, omitting a zero guard term. Within any nonempty signed subset of two or more terms there is a unique least order, and it is even. Grouping by rational squareclass proves that a rational signed sum forces every term rational: within a squareclass, the unique least valuation cannot cancel. This is the retained prime-order separation lemma, now applied before any positivity of D_g is known. In particular D_g is a nonnegative integer square.

Since $s \neq 0$, $B_0 \leq 0$ would give $D_g \leq -3$. Therefore $B_0 > 0$. The elementary joint-positivity lemma for $xy - (t^2 - x - y + 1)^2$ gives

$$G > 0, \qquad \eta\zeta > 0. \tag{23}$$

If $c = 0$, then $\eta\zeta = -Y(Y + 1)k^2 \leq 0$, a contradiction. Thus $c \neq 0$, $I \neq 0$, and now $Q > 1$. Adjacent-coefficient Pell descent makes the rational auxiliary ordinate integral in

$$QV^2 - (Q - 1)y^2 = 1. \tag{24}$$

In the integral presentation one first puts $y = \sqrt{\delta C_y}/\delta$. The other three extracted roots are integers f, τ, g. This order handles the $c = 0$ and $Q = 0$ branches without prematurely invoking a descent theorem that requires $Q > 1$.

5.3 Recover the original scale before the source bounds

The integral signed norm relation gives $S \mid A_*$. Applying Lemma 4.2 with $L = v - q^2$ yields

$$2q^2 \mid v - q^2, \qquad H \mid K_*^5.$$

There is consequently an integer u with

$$v = q^2(2u+1), \qquad Y = q^3(2u+1). \tag{25}$$

The full original large-scale source front is restored here, before its signed Pell estimates, rank extraction or native decoding is used. This also excludes $v = 0$ by an actual argument.

Let $K = X(X + 2ac) - 1$. The exact identity

$$K_* = 2K - HXc \tag{26}$$

and oddness of H give $H \mid K^5$. The short projector itself is already present in the repository's positive-domain work; the new use is its signed transfer into this weaker mixed-modulus carrier.

5.4 Fold the main Pell root, then use the stronger height

The retained signed main-folding argument uses the strong equation $f^2 - \Delta I^2 = 1$, the auxiliary equation (24), the bound $|c| \le |I|$ and $V \equiv -c \pmod f$. It gives $|c| = \psi_{|a+2|}(p)$. The $c^2 \mid I$ rank argument and the two auxiliary residues then give $p = R$. These steps precede any main projection. Their existing ratio bound supplies

$$R = 2r_0 + 1, \quad r_0 \ge 24, \quad x = |X|, \quad m = |Y| > \frac{(x-3)^{r_0}}{2} - 1. \tag{27}$$

With $\sigma = \operatorname{sgn}(c)$, the same projection identity gives

$$2^R K \equiv B := (\sigma X - 2^R)(2^R \sigma X + 1) \pmod H.$$

Thus $H \mid B^5$.

Lemma 5.1 (Height at the actual coefficient tuple)**.** *Under* (21) *and* (27)*,* $|B|^5 < |H|$.

Proof. Each factor of B has absolute value at most $2^{R+1}x$, whence $|B|^5 \le 2^{20r_0+20}x^{10}$. Put $L = (x-3)^{r_0}$. Since $L \ge 4$, $m > L/4$, and

$$|H| \ge 4(x-1)m - 3 > (x-1)L - 3 \ge \frac{x}{2}L \ge \left(\frac{x}{2}\right)^{r_0+1}.$$

Finally $x^{r_0-9} \ge 2^{36r_0-324} > 2^{21r_0+21}$ for $r_0 \ge 24$. This is exactly the remaining comparison. □

Lemma 4.1 gives $B = 0$. Its second factor is odd and nonzero, so $\sigma X = 2^R$. The ordinary main projection and native decoding now recover a finite refutation. This proves soundness. The stronger estimate depends on the actual large β; it is not a claim valid under only the earlier generic bound $x \ge 64$.

6 Completeness and the order of the witnesses

Begin with a finite refutation and its canonical source data. Write the old cubic-scale parameter as u, and set $v = q^2(2u+1)$. All quantities depending on Y are then unchanged. The canonical projection gives $H \mid K$, hence $H \mid K_*$. Since $M \mid v - q^2$,

$$\kappa = C_*/D = K_*/H + (v - q^2)/M$$

is an integer.

Choose the joint guard witness t and its nonzero Pell witness s. The retained full-residue interpolation theorem then chooses one strong pair I, f with $4sc^2 \mid I$. Put $r = I/(sc^2)$. *Fix this pair before choosing the residue of* T. Intersectivity gives a root u_0 of π modulo I. Request $T \equiv u_0 - \kappa \pmod{I}$; interpolation supplies such an auxiliary witness without changing I. Then

$$A_* = D^5\pi(T+\kappa), \qquad \frac{A_*}{S} = \frac{c^2H^4\pi(T+\kappa)}{I} \in \mathbb{Z}. \tag{28}$$

For (18) take

$$n = A_*/S + 3f + 9\tau + 27g + y - V.$$

For (19) replace $y - V$ by $\delta(y - V)$. A conjugate norm factor vanishes. Every coordinate is integral; n has no positivity restriction. The witness order is outer data, t, s, I, f, an intersective residue, T, then n. In particular there is no circular choice of a modulus constrained by a later residue.

Theorem 6.1. *The two equations* (18) *and* (19) *have integer zeros exactly when the pinned source has a finite refutation. Composing with the existing source/calculus calibration proves* (1) *for both equations. Their exact degrees are* 768 *and* 784.

7 Centering, degree certificates, and the cheaper variants

For P_{768} let $U = Sn - A_*$, $P = SV$, $L = \delta P$, and $N = S^2 - P^2$. Pairing the two signs of the auxiliary radical gives

$$\delta W_1^2 + 2LW_1 + N, \qquad W_1 = U - 3Sf - 9S\tau - 27Sg. \tag{29}$$

Taking the remaining three-root norm is exactly the cleared original norm. For P_{784} put $L = S\delta V$ and $N = S^2\delta - L(SV)$; the paired expression is

$$W_1^2 + 2LW_1 + N. \tag{30}$$

These are universal ring identities. They remove the high terms before degree propagation and correspond to the integral coordinate changes $n \mapsto n + V$ or $n \mapsto n + \delta V$.

The relevant degrees are

$$\deg H = 4,\ \deg M = 2,\ \deg D = 6,\ \deg I = 4,\ \deg \delta = 24,\ \deg V = 21,\ \deg S = 16,\ \deg A_* = 35.$$

In (29), $2LU$ has degree 96, above the competing 94 and 74. Its eight conjugates give degree 768. In (30), N has degree 98, above the degree-96 mixed term, giving 784. The exact nonzero coefficients

$$[j^{352}w^{96}v^{96}s^{48}r^{48}c^{80}T^{48}]P_{768} = 2^{744}3^{448}\beta^{352}, \tag{31}$$

$$[j^{320}w^{80}v^{80}s^{80}r^{80}c^{128}T^{16}]P_{784} = 2^{624}3^{400}\beta^{320} \tag{32}$$

prove attainment at the actual coefficients. The checker independently derives the leading term at every live gate over all six formal ports; the coefficients in (31)–(32) are strictly positive after specialization.

For the cheaper variants, retain the original $Y = (2v+1)q^3$, use $D = H$, $z_* = HT + K_*$ and $S = srH$, and use the same Φ_H. The guard-first $c \neq 0$ argument is unchanged. The carrier forces $H \mid K^5$, and Lemma 5.1 resolves it. Without the mixed-modulus circuitry, the integral and rational forms have respectively degrees $1040,816$ in $140,144$ operations. Centering the integral form reduces its degree to 800 in 147 operations. Its attained coefficient is

$$[j^{384}w^{96}v^{96}s^{48}r^{48}c^{80}T^{48}]P_{800} = 2^{792}3^{480}\beta^{384} > 0.$$

Their complete proofs, graphs and checks are in `height12/`; the mixed-modulus forms are in `crt12/`.

8 A second construction: combine signs of bounded intervals

The direct branch starts from the pinned 75-predicate raw C222 system [6]. Its body has 1259 coordinates and 1177 quadratic residuals. The first 594 rows form a front encoding CRT, factorial, binomial and flow data. Power requests in that front are certified by shared Pell witnesses. The frozen source has an ordinary proof of its finite-refutation equivalence, independent of K215 and the native tuple.

The direct logical endpoint is precise. The inherited `KEndpoint.c222_iff_zf_refutation` theorem identifies $C_{222}(1)$ with a finite proof of contradiction from ZF. The inherited `ConstructibleChoice.zf_refutation_iff_zfc_refutation` theorem identifies that with a finite ZFC refutation. The latter is the ordinary constructible-universe equiconsistency theorem, without an added consistency assumption. `Bridge.lean` defines `ConZFC` as the negation of that exact ZFC proof predicate. These unconditional calibration theorems, their original pins and the intervening ordinary arithmetic proofs are included in `inherited/direct/` and the immutable incoming archive. Their Lean import closure has not been recompiled in this work. The later machine-specific endpoint in the same upstream file is not needed by the direct construction.

Lemma 8.1 (Shared signs)**.** *If $r + s = a + b$ and $a, b \geq 0$, then*

$$(r \geq 0 \ \wedge \ s \geq 0) \quad \longleftrightarrow \quad rs \geq 0.$$

Proof. If $r < 0$, then $s \geq -r > 0$, so $rs < 0$; interchange r, s for the other case. The forward implication is immediate. This includes zeros. □

For an integer x, Legendre's three-square theorem gives

$$x \geq 0 \quad \longleftrightarrow \quad \exists a, b, c \in \mathbb{Z} \quad x = a^2 + b^2 + c(c+1). \tag{33}$$

Indeed $4x + 1$ is representable by three squares; modulo four exactly one root is odd. Conversely $c(c+1) \geq 0$ for every integer c. This is also the explicit sign representation recorded in [8].

Replacing two scalar sign conditions by one product condition therefore uses one quadratic row and three auxiliary integers instead of two rows and six. The actual seven replacements are shown below. Indices are zero-based in the unchanged raw body. Each listed three-row range sums exactly to $Q - r - s - 1$.

Product	Nonnegative anchor pair	Common Q	Product / anchor rows
$x_{562}x_{563}$	(x_{549}, x_{550})	x_{547}	537–539 / 526–528
$x_{585}x_{586}$	(x_{576}, x_{577})	x_{574}	553–555 / 547–549
$x_{593}x_{594}$	(x_{576}, x_{577})	x_{574}	559–561 / 547–549
$x_{601}x_{602}$	(x_{576}, x_{577})	x_{574}	565–567 / 547–549
$x_{619}x_{620}$	(x_{610}, x_{611})	x_{608}	578–580 / 572–574
$x_{637}x_{638}$	(x_{628}, x_{629})	x_{626}	591–593 / 585–587
$x_{123}x_{564}$	(x_{91}, x_{537})	x_{530}	540–542 / 518–520

The first six anchors retain both scalar guards. For the last, rows 83–85 give $x_{91} = x_{94} + x_{95} + 1$, with x_{94}, x_{95}, x_{537} still guarded. All fourteen deleted signs are thus recovered *before* any Pell, power or CRT semantics is invoked. The parent body assignment is unchanged in both directions; only private sign witnesses are replaced.

This yields 68 predicates. After complete integral monic projection it gives 589 variables, 371 quadratic rows and 8157 collected quartic terms. The projected gain is 15 variables, not the raw 21-witness saving: six product constraints prevent six formerly available coordinate eliminations. This is why actual generated polynomials, rather than subtraction of guard counts, are used for every size claim.

9 Even indices remove the remaining small-ordinate guards

The next change is structural. Instead of certifying $m = n^k$, certify

$$m^2 = n^{2k}. \tag{34}$$

The front already supplies $m \geq 0$, $n \geq 1$, and $k \geq 1$, so (34) is equivalent to the original equality. Squaring an output in its quadratic congruence row preserves degree two; doubling an index changes only a scalar coefficient. The benefit is that every requested Pell index is now even.

9.1 Pell notation and a residue lemma

For $a \geq 2$ write

$$(a + \sqrt{a^2 - 1})^j = \chi_a(j) + \psi_a(j)\sqrt{a^2 - 1}.$$

Both sequences are integral, χ_a is strictly increasing on nonnegative indices, and $\psi_a(j) \geq j$. Standard reduction by the fundamental norm-one unit $a + \sqrt{a^2 - 1} > 1$ classifies every integral Pell solution, up to independent signs, by these sequences. For a signed base B, define χ_B, ψ_B by the same integer-polynomial recurrences. For $b > 0$ they satisfy $\chi_{-b}(j) = (-1)^j \chi_b(j)$ and $\psi_{-b}(j) = (-1)^{j-1}\psi_b(j)$ for $j \geq 1$.

Lemma 9.1 (Abscissa residues)**.** *If $1 \leq i \leq h$ and $\chi_a(j) \equiv \pm\chi_a(i) \pmod{\chi_a(h)}$, then $j \equiv \pm i \pmod{2h}$. In particular $\chi_a(j) \equiv \pm a \pmod{\chi_a(h)}$ forces j odd.*

Proof. The doubling formulas give $(a + \sqrt{a^2 - 1})^{2h} \equiv -1 \pmod{\chi_a(h)}$ in the quadratic ring. Reducing an index modulo $2h$ and reflecting about h reduces its abscissa to $\pm\chi_a(r)$ with $0 \leq r \leq h$. For $h \geq 2$, the recurrence implies $2\chi_a(h-1) < \chi_a(h)$. Two abscissae with indices below h cannot sum to the modulus, so congruence up to sign forces equality, hence $r = i$. If $i = h$, only the endpoint has zero residue. The case $h = 1$ is immediate. Taking $i = 1$ proves the final assertion. □

9.2 The new shared frame

There are 21 calls and 15 distinct seed indices. Let $\mathcal{O}$ be the 21 output coordinates and $\mathcal{F}$ the frozen set of 43 distinct frame arguments. Define

$$W = 4 + 2 \sum_{v \in \mathcal{F} \setminus \mathcal{O}} v + 2 \sum_{m \in \mathcal{O}} m^2. \tag{35}$$

Every summand is nonnegative before power interpretation. Consequently $W \geq 4$, and every call has $n < W$, $2k < W$, and $m^2 < W$. The unchanged frame equation is

$$A^2 - W^3(W+2)Z^2 = 1.$$

The retained first modulus guard forces $A \geq 2$, hence $Z \neq 0$. Classifying this equation at base $W+1$, and reducing its ordinate modulo W, shows the frame index is a positive multiple of W. Thus

$$A \geq (W+1)^W > W^W > W^2. \tag{36}$$

For every base $n \geq 1$, the unchanged modulus $T_n = 2An - n^2 - 1$ satisfies $T_n > A$.

9.3 The parity argument, including arbitrary signs

The common pair satisfies

$$U^2 - (A^2-1)V^2 = 1, \quad V \geq 1, \quad V = \Big(\prod_{i=1}^{15} Y_i^2\Big)(v_0+1), \quad B = 1 + 4V(q_0+1), \quad B \equiv A \pmod{U}. \tag{37}$$

Here v_0, q_0 are the common-certificate coordinates, distinct from the twelve-variable branch's notation. Every Y_i is nonzero and $Y_i^2 \mid V$ by the literal product identity. The odd B cannot equal ± 1: with $|U| = \chi_A(h)$, the residue $B \equiv A \pmod{U}$ excludes both possibilities by size, including $h = 1$.

For a seed with requested even index $K = 2k$, retain

$$X^2 - (A^2-1)Y^2 = 1, \qquad S_0^2 - (B^2-1)R_0^2 = 1, \tag{38}$$

$$S_0 \equiv X \pmod{U}, \qquad R_0 \equiv K \pmod{4Y}. \tag{39}$$

If $|Y| = 1$, then $X = \pm A$. Writing the auxiliary solution with arbitrary signs and transferring the base modulo U gives $\chi_A(j) \equiv \pm A \pmod{\chi_A(h)}$. Lemma 9.1 forces j odd. Since B is odd, $\psi_B(j) \equiv j \pmod 2$, so R_0 is odd. But its residue modulo $4Y$ forces it even. This contradiction excludes $|Y| = 1$.

Therefore the main index is at least two, and

$$|Y| \geq \psi_A(2) = 2A > K. \tag{40}$$

The old index-slack nonnegativity guards are no longer needed to obtain this strict bound. The argument allows negative U, B, X, Y, S_0, R_0.

9.4 The index is exactly the requested one

Suppose $|X| = \chi_A(i)$, $|Y| = \psi_A(i)$ and $V = \psi_A(h)$. The elementary squared-divisibility identity says

$$\psi_A(i)^2 \mid \psi_A(h) \quad \Longrightarrow \quad i \leq h \ \text{ and } \ \psi_A(i) \mid h.$$

For completeness, write $h = qi + r$ with $0 \leq r < i$. Reduction modulo $\psi_A(i)$ and strict monotonicity give $r = 0$; binomial expansion modulo $\psi_A(i)^2$ then gives $\psi_A(i) \mid q$, hence the claim.

The abscissa residue gives $j \equiv \pm i \pmod{2h}$. Since $B \equiv 1 \pmod{4|Y|}$, the ordinate residue gives $j \equiv \pm K \pmod{4|Y|}$. As $|Y| \mid h$, these imply $i \equiv \pm K \pmod{2|Y|}$. With $1 \leq i \leq |Y|$ and $1 \leq K < |Y|$, only $i = K$ is possible.

9.5 Orient seed pairs before allowing base one

There are designated supporting calls for all fifteen seeds, with indices

$$0, 1, 5, 7, 9, 10, 12, 13, 14, 15, 16, 17, 18, 19, 20.$$

Their bases are already at least two from the front, before any factorial or power meaning is used. This qualification matters: the base $x_{112} = 1 + x_{107}x_4^2$ of another call can initially be one.

For a supporting call put $e = 2k$, $N = n^e$, $M_0 = m^2$, and $x = \chi_A(e), y = \psi_A(e)$. With $H_0 = x-(A-n)y$, $J_0 = x + (A - n)y$, the recurrence gives

$$H_0 \equiv N \pmod{T_n}, \qquad H_0 J_0 = 1 + T_n y^2.$$

The four possible signs of (X, Y) in the modified power congruence give respectively

Signs	Consequence
$(+,+)$	$M_0 \equiv N \pmod{T_n}$
$(-,-)$	$T_n \mid M_0 + N$
$(+,-)$	$T_n \mid M_0 N - 1$
$(-,+)$	$T_n \mid M_0 N + 1$

The frame gives $M_0 N < W^W < A < T_n$ and $M_0 + N < W + W^{W-1} < W^W$. The last three cases are impossible; in particular $M_0 N = 1$ is impossible because $n \geq 2, e \geq 2$. Thus every seed pair is positively oriented and $M_0 = N$. Nonnegativity gives $m = n^k$. After this orientation, the remaining calls, including those with base one, use the positive pair and the simple bounded congruence; no inverse-residue sign ambiguity remains.

9.6 One derived pair needs two steps

The old $t + 1$ pair was computed by one Pell step from the t seed. After doubling, the seed index is $2t$ and the target is $2(t + 1)$. Thus two steps are essential. With stored coordinates for AX, $(A^2 - 1)Y$ and A^2, the replacements are

$$X_{+2} = 2A(AX + (A^2 - 1)Y) - X, \tag{41}$$

$$Y_{+2} = 2AX + (2A^2 - 1)Y. \tag{42}$$

They are quadratic in the stored coordinates, so no higher-degree residual is introduced. Omitting this change would invalidate the compiler.

10 Completeness and the literal quartic

Given true front arithmetic data, compute the new W. Choose

$$A = \chi_{W+1}(W), \qquad Z = \psi_{W+1}(W)/W.$$

The latter is integral by the recurrence modulo W. Choose positive main pairs at every $2k$ and the derived pair by two steps. Let $P = \prod_{i=1}^{15} Y_i^2$ and use the finite ring

$$(\mathbb{Z}/4P\mathbb{Z})[u]/(u^2 - (A^2 - 1)).$$

The unit $A+u$ has finite multiplicative order. A positive multiple of that order gives a common pair with $U \equiv 1 \pmod{4P}$ and $V \equiv 0 \pmod{4P}$, so $V > 0$, $P \mid V$ and U is odd. As $\gcd(U, 4V) = 1$, CRT gives $B \equiv A \pmod{U}$ and $B \equiv 1 \pmod{4V}$. Choose $B > \max(A, 1+4V)$ in that progression; monotonicity and the recurrences give nonnegative comparison quotients. Choose the auxiliary pairs at the same $2k$. Polynomial dependence on the base gives both required congruences.

The power quotient is supplied by the exact recurrence identity

$$\chi_A(e) - (A-n)\psi_A(e) - n^e = (2An - n^2 - 1)\sum_{j=0}^{e-1} n^{e-1-j}\psi_A(j). \quad (43)$$

Thus all twenty-one modified power rows hold. This is a simultaneous construction with the explicit order front, frame, main pairs, common pair, CRT base, auxiliary pairs, quotients. No incompatible preselected auxiliary witnesses are reused.

From a new zero, the preceding soundness proof recovers the original twenty-one equalities at the same front tuple. Build a *fresh old* frame and certificate at indices k, using one step for its derived $t+1$ pair, to obtain a zero of the pinned parent. Conversely, old soundness supplies true front data, from which the new completeness construction produces a zero. The maps concern zero existence, not pointwise equality of old and new auxiliary coordinates.

The exact source edits consist of fifteen doubled seed residues, twenty-one squared outputs, twenty-one squared frame contributions, two derived-pair rows, and the doubled frame endpoint. Eleven scalar index guards and one paired-index guard are deleted, leaving 56 predicates after sign sharing. The complete source edit ledger and all projection restorations are supplied in `direct566/`.

Let $q_1, \ldots, q_{372}$ be the literal quadratic residuals in its `equation.json`, in the stated order of 566 integer coordinates. The second principal equation of this paper is

$$\boxed{Q_{566}(x_0, \ldots, x_{565}) = \sum_{i=1}^{372} q_i(x_0, \ldots, x_{565})^2 = 0.} \quad (44)$$

The accompanying sparse coefficient file lists every one of the 6431 nonzero monomials. All fixed coefficients are literal integers, and no runtime call to a proof predicate or to a universal polynomial remains. An integer zero of (44) is exactly a common zero of those rows.

Theorem 10.1. *The complete polynomial Q_{566} has degree four,* 566 *unrestricted integer variables,* 6431 *monomials, and coefficient height* 2445790974704. *Its integer-zero existence is equivalent to that of the pinned direct parent and hence, using the existing direct C222/calculus calibration, to finite ZFC refutability.*

11 Verification, limitations, and further research

11.1 What the checks establish

The package includes independent checkers, ordinary proofs and full input data. Verification is deliberately split into mathematical and finite algebraic claims:

- The twelve-variable checkers authenticate all six payloads, prove every named register by exact free-ring identities, verify the complete 495-term four-root norm and centered cuts, check acyclicity and liveness, and derive attained leading coefficients through the actual DAGs.

- The direct checkers bind the seven sign replacements to literal source-row identities, bind every changed power/index/frame row, verify the two-step correction and the actual fifteen-factor common product, restore every projected coordinate by an integral polynomial, and expand the entire final sum of squares independently.
- Normal and optimized Python execute the same checks. Deliberately changed projectors, carrier operations, source products, restorations and coefficients are rejected. No mathematical acceptance depends on Python assertions that disappear under optimization.
- The degree and coefficient counts are exact. Finite Pell examples and the negative control are supplementary diagnostics; the unbounded proofs above establish the equivalences.

The supplied odd-index negative control shows why merely deleting the index guards is invalid. It gives a genuine frame and a single generic power certificate with $n = 4, k = 5, m = 2$, $|Y| = 1$, satisfying the squared-output condition while $m \neq n^k$. The doubled-index residue rejects it. This is a counterexample to a generic shortcut, not a root of the ZFC equation.

11.2 Precise boundaries

The words "ZFC-provable" describe an ordinary mathematical proof that can be formalized inside ZFC, composed with the explicitly cited source theorems. They do not claim that this package emits a new Lean/Rocq proof of the entire native or direct compiler chain. The inherited calibration and the new local algebraic certificates have different scopes and remain separately identified. None of the work assumes ZFC consistency, and no root search can establish the desired absence of roots.

The elementary ingredients are classical. The claimed progress is the specific changes to the fully instantiated constructions at the pinned snapshot, with exact new equations and verified comparisons. We do not claim these are globally simplest, nor that the local lemmas have no precedent anywhere in the literature.

11.3 Further research questions

1. Can more of the actual coefficient-size margin be exchanged for weaker carrier divisibility, while preserving both directions of the source reconstruction?
2. Can the M^5 condition be shortened by a proved restriction on the prime factors of a canonical strong modulus? Generic modular roots do not suffice: primes dividing the mandatory c and s must be covered.
3. The scale $Y = qv$ retains the uniform coprimality $\gcd(H, 2q^2) = 1$. Dropping the last factor q would reduce degree further but breaks that uniform fact. Can a different carrier repair it without a new variable or a circular coprimality assumption?
4. Can the even-index compiler be applied before the current raw arithmetic expansion, so its source frame and shared certificates can be redesigned together rather than locally rewritten?
5. Can integral changes of coordinates recover the six eliminations lost when two scalar guards become a product guard? Can a larger row basis expose additional independently oriented sign pairs?
6. Can the new transformations be formalized once as reusable theorems, then applied to an authenticated complete native or direct source theorem? The proof order around zero scale, base one, derived indices and common periods should be explicit in such a formalization.

A A division-free norm evaluator

This appendix makes explicit that the displayed norm is an ordinary polynomial. Let the desired cleared norm be

$$d^8\mathcal{N}_4(x; a, b, e, c/d).$$

For the integral norm set $d = 1$. Calculate in the quadratic algebra with a formal symbol u satisfying $u^2 = e$. Put

$$\begin{aligned}
t_{20} &= x^2 + e, & t_{21} &= -2x, & A &= a + b,\\
q_0 &= d(t_{20} - A) - c, & q_1 &= dt_{21}, & &\\
T &= d^2ab, & R &= dc, & &\\
v_0 &= q_0^2 + eq_1^2 - 4(T + RA), & v_1 &= 2q_0q_1, & &\\
Z &= 64TR, & & & &\\
n_0 &= v_0^2 + ev_1^2 - Zt_{20}, & n_1 &= 2v_0v_1 - Zt_{21}. & &
\end{aligned}$$

The result is $n_0^2 - en_1^2$. Every operation is in the integer polynomial ring. Expanding the product of all sixteen conjugates verifies the identity with 495 nonzero formal coefficients.

For the centered rational form, use $x = U + P$, $L = dP$, $N = S^2 - P^2$, and $c = dP^2 - N$. Replace the expression for q_0 by

$$q_0 = U(dU + 2L) + N + d(e - a - b).$$

For the centered integral form, use $x = U + L$, $c = L^2 - N$, $d = 1$, and

$$q_0 = U(U + 2L) + N + e - a - b.$$

These local identities are checked separately before the full norm identity is used. They explain the lower attained degrees without any assumption that a residual has already vanished.

B Package map and reproduction

The main deliverables are this PDF and its complete LaTeX source, together with the following directories:

Path	Contents
`crt12/`	Degrees 768/784 literal DAGs, builder, independent checker, complete coefficient payloads and full carrier proof.
`height12/`	Degrees 800/816/1040 graphs and ordinary height/centering proofs, with independent exact checks.
`direct566/`	Complete even-index quartic, quadratic residuals, expanded monomials, source edits, projection and independent checks.
`direct589/`	The separately verified seven-sign-combination intermediate and lower-height alternatives.
`peer/`	Independent mathematical and exact circuit reviews, with the generic negative control in `direct566/`.
`inherited/`	Pinned ordinary proof dependencies, baseline review and the original direct calibration archive.
`verify_all.py`	One entry point for all new checks and deterministic circuit reproduction.
`SOURCE_LEDGER.json`	Pinned upstream identities, source locations, and clear inherited-versus-new proof scope.

From the extracted package root run

```
python3 verify_all.py
python3 -O verify_all.py
pdflatex -interaction=nonstopmode -halt-on-error report.tex
pdflatex -interaction=nonstopmode -halt-on-error report.tex
```

The numerical verifiers use Python's standard library. The parent coefficient recipes and full logical calibration are pinned to upstream; the actual coefficients and all new final polynomials are included here.

References

[1] V. Reshetnikov, *ProveIt*, snapshot `84e16e43092614bae262ca59651cb9d77e31a255`, 24 October 2026. https://github.com/VladimirReshetnikov/ProveIt/commit/84e16e43092614bae262ca59651cb9d77e31a255.

[2] ProveIt, *Later arrivals and current size reductions*, `Computability/DiophantineConsistency/PROPOSAL_REVIEW_LATEST.md`, at the preceding snapshot. This is the baseline for the comparison table.

[3] ProveIt, *Scalar intersective carrier after signed main-root recovery*, `Computability/DiophantineConsistency/candidates/scalar-carrier12/PROOF.md`, same snapshot; complete coefficient binding and graphs in that directory.

[4] ProveIt, `candidates/scalar-carrier12/source/PRIME_TWELVE.md`, within `Computability/DiophantineConsistency`, same snapshot.

[5] ProveIt, `candidates/scalar-carrier12/source/INTERPOLATION12.md`, same snapshot.

[6] ProveIt, `research/factorial-radius-output-sign-profile/raw/parent.json` and `raw/domain.json`, within `Computability/DiophantineConsistency`, same snapshot; associated ordinary proofs and the active review identify the direct C222, shared-CRT and Pell calibration.

[7] ProveIt incoming package, `ConZFC_407_888_Quartics_2026-10-24.zip`, same snapshot. The unchanged terminal-elimination engine is used as a producer; independent checkers validate its complete integral restoration certificates.

[8] Z.-W. Sun, *Ten unknowns for Hilbert's tenth problem over the integers*, arXiv:2609.01594v2, 2 September 2026. https://arxiv.org/abs/2609.01594v2.